

`KNeighborsClassifier` (from `sklearn.neighbors`) inherits from several base classes, so it exposes both its own methods and common scikit-learn estimator methods.

Below is a **practical list of the main public methods** you can use. (Exact availability can vary slightly by scikit-learn version.)

Core estimator methods

- `fit(X, y)` – Fit the k-NN classifier from training data
 - `predict(X)` – Predict class labels for samples in `x`
 - `predict_proba(X)` – Predict class probabilities for samples in `x`
 - `score(X, y)` – Return mean accuracy on the given test data and labels
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Neighbor search–specific methods

- `kneighbors(X=None, n_neighbors=None, return_distance=True)`
Find the K nearest neighbors of points
 - `kneighbors_graph(X=None, n_neighbors=None, mode='connectivity')`
Compute the k-neighbors graph for points in `x`
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Scikit-learn estimator utility methods

- `get_params(deep=True)` – Get parameters for this estimator
 - `set_params(**params)` – Set the parameters of this estimator
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Validation / metadata helpers (newer versions)

- `get_metadata_routing()`
- `set_fit_request(...)`
- `set_predict_request(...)`
- `set_score_request(...)`
- `set_params(...)`

(These support scikit-learn's metadata routing system.)

Common learned attributes (after `fit`)

Not methods, but frequently used:

- `classes_` – Class labels known to the classifier
 - `effective_metric_`
 - `effective_metric_params_`
 - `n_features_in_`
 - `feature_names_in_` (if input had feature names)
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Quick way to see them yourself

```
from sklearn.neighbors import KNeighborsClassifier
clf = KNeighborsClassifier()
[m for m in dir(clf) if not m.startswith("_")]
```